

NAME: _____

PERIOD: _____

DATE: _____

Which Summary Tells the Story?

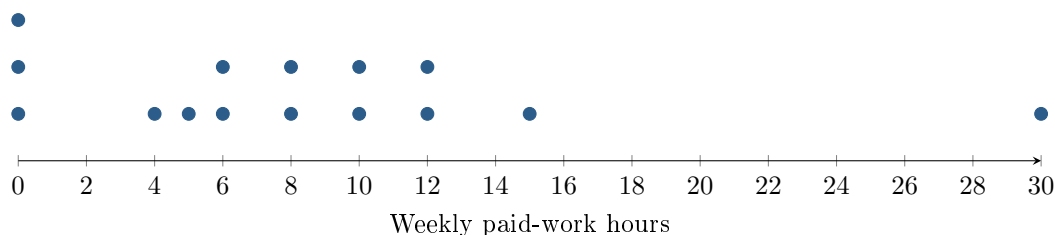
AP Statistics · Unit 1 · Topic 1.7 · B: 2026-09-21 · E: 2026-09-23

[STAR] TODAY'S PROBLEM

Suppose a principal asks, “How much paid work do students do in a week?” Fifteen students report these weekly hours: 0, 0, 0, 4, 5, 6, 6, 8, 8, 10, 12, 12, 15, 30.

Kai: “ $\bar{x} = 8.4$, $s_x \approx 7.5$, and $\text{IQR} = 8$. Report the mean and SD because they use every observation.”

Rosa: “Median = 8, $Q_1 = 4.5$, $Q_3 = 11$, and $\text{IQR} = 6.5$. Report the median and IQR because 30 looks unusual.”



FIRST TAKE — before we discuss (5 min)

Whose report would you send, Kai's or Rosa's? Cite one number or dotplot feature.

[BOOK] CENTER, SPREAD, AND OUTLIERS (keep on the page)

The median is the middle ordered value (average the two middle values for an even count). For this sheet, find Q_1 and Q_3 as the medians of the lower and upper halves, excluding the overall median when n is odd. $\text{IQR} = Q_3 - Q_1$. Outlier fences are $Q_1 - 1.5\text{IQR}$ and $Q_3 + 1.5\text{IQR}$. Mean and SD are nonresistant: extremes affect them more. Median and IQR are resistant. Use the distribution's shape to choose a pair.

Work (a)–(c) from the rules box:

DOK 1 (a) State the median and maximum weekly paid-work hours.

DOK 2 (b) Using the printed quartile rule, find Q_1 , Q_3 , the IQR, and the upper 1.5-IQR fence. Is 30 beyond it? Use Kai's mean and SD as given.

DOK 3 (c) Whose IQR follows the printed rule, and how did the other result arise? Choose a corrected summary pair for the principal. Use a dotplot feature to explain why that pair describes typical work hours better than the alternative.

Frame: _____'s IQR follows the printed rule; the other calculation includes _____.

Frame: I would report _____ because the dotplot shows _____; the alternative could make the principal believe _____.

EXIT — turn this in

One thing the rules box changed about my first take: _____